

Fisher Information and Coupling Reconstruction from Edge-Sector Statistics in Two-Dimensional Yang–Mills

William T. Trevena

Independent Researcher (PhD, ISE, University of Florida)

trevenaw7@gmail.com

June 2026

Acknowledgment: Portions of this manuscript were developed with the assistance of Claude (Anthropic), an AI language model, which contributed to literature synthesis, mathematical exposition, drafting, and the computational experiments. Every formula taken from the literature was verified against the arXiv LaTeX source of the cited paper, not from memory; the verification record is `ym2/FORMULAS.md` in the repository. All numerical claims are reproduced by the published scripts (`ym2/ym2_flux.py`, `ym2/ym2_su3.py`, deterministic seed 20260609). Consistent with COPE authorship guidelines, the AI system is acknowledged as a tool rather than listed as a coauthor.

Abstract

The edge-sector distribution of two-dimensional Yang–Mills theory is an exponential family; therefore its entropy derivative, its Fisher information, and the sensitivity of coupling estimation are exactly linked. Concretely: for a nontrivial compact connected Lie gauge group G in the standard heat-kernel formulation on a sphere, the Hartle–Hawking state assigns to the boundary electric flux R at an entangling cut the probability $p_R(t) \propto (\dim R)^2 e^{-(t/2)C_2(R)}$, $t = g^2 A$ — a one-parameter exponential family with natural parameter $-t/2$ and sufficient statistic the quadratic Casimir. The formulas themselves are known (Donnelly; Gromov–Santos); our contribution is the estimation-theoretic reading of them, stated as theorems with explicit hypotheses, plus the explicit $SU(3)/\text{capacity}/n$ -interval analysis. Three exact statements result: (i) the entanglement entropy obeys $dS/dt = -(t/4) \text{Var}_t(C_2) < 0$; (ii) the Fisher information of the flux measurement is $I(t) = \frac{1}{4} \text{Var}_t(C_2)$, and it equals the quantum Fisher information, so $dS/dt = -tI(t)$: entropy susceptibility and statistical reconstructability are the same function of the coupling; (iii) the capacity of entanglement is $C(t) = t^2 I(t) \rightarrow \frac{1}{2} \dim G$ as $t \rightarrow 0$, and $S(t) = \frac{1}{2} \dim G \ln(1/t) + c_G + o(1)$ with c_G the heat-kernel constant computed in the appendix. The operational consequence is statistical: an observer who samples the boundary flux N times estimates t at the Cramér–Rao limit — for $U(1)$, $SU(2)$, and $SU(3)$, the coupling is recovered to 1.6% from 10^3 samples and 0.5% from 10^4 at $t = 1$, with finite-sample bias matching the analytic $O(1/N)$ formula (reproducibility appendix). Exact inversion of the convention-fixed entropy is the *idealized equivalent* of that sampling channel, not an independent protocol: entropy is not an observable, flux-sector sampling is the physical channel, and the Fisher identity is why the two sensitivities coincide. With n intervals the entropy grows linearly in n through the edge term $2n \langle \ln \dim R \rangle$, but the Fisher information does not grow at all: the flux at every cut is one global random variable, so entropy scales while information does not. We state precisely why all of this is *reconstruction of a parameter within a presupposed theory, not determination of the coupling by entanglement*: the inference inverts a known one-parameter family; only the product $g^2 A$ is recoverable; the edge term that carries the signal is contested as entanglement, so the defensible reading is reconstruction from superselection-sector statistics; and no claim is made about four-dimensional QCD. We close with a checklist that any future “coupling from entanglement” claim should specify.

1 Introduction

1.1 The observation and the identity chain

Pure Yang–Mills theory in two spacetime dimensions on a sphere is exactly solvable, and the gauge-invariant data at an entangling cut of the Hartle–Hawking state is a single discrete random variable: the boundary electric flux, an irreducible representation R of the gauge group, distributed as

$$p_R(t) \propto (\dim R)^2 e^{-(t/2) C_2(R)}, \quad t = g^2 A.$$

The observation that drives this paper is elementary but, as far as we could determine, unexploited: **this edge-sector distribution is a one-parameter exponential family**, with natural parameter $-t/2$ and sufficient statistic the quadratic Casimir $C_2(R)$. Everything we prove is a corollary of that single fact. The entanglement entropy of the cut, the Fisher information of the flux measurement, and the sensitivity of any estimate of the coupling are exactly linked:

$$\frac{dS}{dt} = -\frac{t}{4} \text{Var}_t(C_2) = -t I(t) = -\frac{C(t)}{t},$$

where $I(t)$ is the (classical = quantum) Fisher information about t of one flux sample and $C(t)$ is the capacity of entanglement. Entropy susceptibility, flux susceptibility, Fisher information, and capacity of entanglement are one function of the coupling, and that function is what makes the coupling statistically recoverable.

The operational channel is sampling. An observer who measures the boundary flux N times estimates t at the Cramér–Rao limit, because maximum-likelihood estimation in a regular one-parameter exponential family is asymptotically efficient; a reproducibility appendix verifies the finite- N behavior. Exact inversion of the strictly monotonic entropy $S(t)$ is the *idealized equivalent* of that channel — entropy is not an observable but a functional of the state — and we keep the two carefully separated throughout (Sec. 3.4).

The input formulas — the heat-kernel solution, the flux distribution, the edge-entropy formula — are known (Donnelly [9]; Gromov–Santos [10]; Sec. 1.5 draws the known/new line item by item). The contribution is the estimation-theoretic reading: the exponential-family structure, the identity chain, the quantum-optimality of the flux measurement, the capacity identity, the n -interval information non-scaling, and the explicit SU(3) analysis.

1.2 Three entropies, and which one this paper uses

Discussions of “entanglement and gauge couplings” move between three distinct objects, and conflating them is the main source of confusion in this area. Table 1 fixes the usage once.

1.3 The honest thesis

Our thesis has a positive half and a deflationary half, and we state both up front.

Positive. In 2d Yang–Mills on a sphere (Hartle–Hawking state, nontrivial compact connected Lie group G , standard heat-kernel formulation), the gauge-invariant edge-sector data of an interval — the distribution $p_R(t)$ of the boundary electric flux, and the entanglement entropy $S(t)$ built from it — determines the dimensionless coupling $t = g^2 A$ **exactly and efficiently**. The operational channel is sampling: an observer who measures the flux N times can estimate t with the smallest variance allowed by statistics, saturating the Cramér–Rao bound, because the flux distribution is an exponential family in the coupling — with no entropy convention needed at all. The idealized

Object	Definition	Status in this paper
Qubit subsystem entropy	von Neumann entropy of the reduced state on a fixed tensor factor (e.g. a qubit)	not used: in gauge systems it is gauge-frame dependent [1]
Algebraic entropy	entropy relative to a gauge-invariant observable algebra (commutant/center construction) [34, 35, 12]	the general definition; specializes to the next row here
Flux-sector (edge) entropy	Shannon entropy of the boundary-flux distribution p_R plus the log dim edge term, eq. (2)	the object of this paper

Table 1: Three inequivalent notions of “entanglement entropy” in gauge systems. This paper works exclusively with the third, which in 2d Yang–Mills coincides with the second: the distillable piece of the algebraic entropy vanishes identically (Sec. 2.2), so the entropy is entirely the edge/sector term. Whether that term is “entanglement” at all is contested, and we keep the caveat prominent (Sec. 5).

channel is inversion: $S(t)$ is strictly decreasing (an identity, not a numerical observation), so with the additive-counterterm convention fixed the map $t \mapsto S$ is a bijection and exact entropy would give t by root finding; but entropy is not an observable, and we present this channel only as the idealized equivalent of the sampling channel, to which the Fisher identity $dS/dt = -t I(t)$ ties it. This includes $G = \text{SU}(3)$: in the $d = 2$ world, the coupling of the $\text{SU}(3)$ theory is reconstructible from edge-sector statistics, and more efficiently than for smaller groups. No claim is made about four-dimensional QCD.

Deflationary. This demonstrates **reflection, not determination**. The inference presupposes the theory — the group, the state preparation, the known one-parameter family $p_R(t)$ — and recovers the value of a parameter the family was already known to depend on. It does not show that the coupling is *constituted by* entanglement, nor that any of this survives to $d = 4$, where the only sharp computation of a coupling-sensitive entanglement coefficient finds it independent of the coupling’s value [18]. Section 5 makes the deflation precise; Section 6 prices out what $d = 3$ and $d = 4$ would require.

1.4 Contributions

Each contribution carries an epistemic label. “Literature-verified” means checked against the arXiv LaTeX source of the cited paper (record: `ym2/FORMULAS.md`). “Derived” means proven here from literature-verified formulas (the proofs are short and given in full). “Computed” means reproduced by the published deterministic scripts.

1. **Monotonicity theorem** (derived; computed to stated residuals). For every nontrivial compact connected Lie group G in the standard heat-kernel formulation, the interval entanglement entropy of the Hartle–Hawking state obeys $dS/dt = -(t/4) \text{Var}_t(C_2) < 0$ on $t \in (0, \infty)$; hence $t \mapsto S(t)$ is a smooth bijection $(0, \infty) \rightarrow (0, \infty)$ and the coupling is recoverable by inversion from the exact *convention-fixed* entropy — equivalently, from the counterterm-free flux distribution $p_R(t)$ (Theorem 3). Numerical inversion recovers t to $\leq 9 \times 10^{-14}$ across three groups (Sec. 3.2).
2. **Fisher-information identity, two reconstruction channels, Cramér–Rao saturation** (derived; computed). The flux distribution is a one-parameter exponential family with

sufficient statistic $C_2(R)$; its Fisher information is $I(t) = \frac{1}{4} \text{Var}_t(C_2)$, so $dS/dt = -t I(t)$ (Theorem 5): the entropy’s sensitivity to the coupling *is* (t times) the statistical information the flux carries about the coupling. This makes the idealized channel (inversion of the exact entropy) and the operational channel (maximum-likelihood estimation from finite flux samples) quantitatively equivalent in sensitivity (Sec. 3.4). The MLE saturates the Cramér–Rao bound for $U(1)$, $SU(2)$, $SU(3)$, with finite-sample bias matching the analytic $O(1/N)$ formula and confidence intervals at nominal coverage (Sec. 4; Appendix B).

3. **$SU(3)$ (computed; new in this paper).** The analysis extends to $SU(3)$ with irreps (p, q) , $\dim = (p+1)(q+1)(p+q+2)/2$, $C_2 = (p^2 + q^2 + pq + 3p + 3q)/3$ (convention verified, Sec. 2.3). At $t = 1$ the coupling is recovered to 1.6% from 10^3 samples, 0.5% from 10^4 , at Cramér–Rao efficiency ~ 1.0 (Appendix B). Reconstruction *improves* with group size: $I(t) \rightarrow \dim G/(2t^2)$ at weak coupling, so the larger the gauge group, the more information each flux sample carries (Sec. 4.1).
4. **Capacity of entanglement** (derived; computed). $C(t) \equiv \text{Var}(-\ln \rho) = t^2 I(t)$, with $C(t) \rightarrow \frac{1}{2} \dim G$ as $t \rightarrow 0$ (verified to 10^{-6} at $t = 0.05$ for all three groups). We found no prior computation of the capacity of entanglement in 2d Yang–Mills (search record in the repository notes); the concept is due to Yao–Qi [25] and was developed in QFT by de Boer–Järvelä–Keski-Vakkuri [26] (Sec. 3.5).
5. **n intervals: entropy scales, information does not** (derived; computed; the qualitative point is Donnelly’s [9]). For a region of n intervals the entropy is $S_n = H[p] + 2n(\ln d)$ [9, eq. (YM2entropy), literature-verified], and we derive $dS_n/dt = -(t/4) \text{Var}(C_2) - (n-1) \text{Cov}(\ln d, C_2)$, strictly negative for all n in all cases computed. But the flux variables at the $2n$ cut points are *perfectly correlated* — one global R — so the Fisher information is exactly n -independent: more cuts give no new information about the coupling, and a numerical experiment confirms the estimator obeys the N -preparation bound, not the hypothetical $2nN$ -sample bound (Sec. 3.6). 2d Yang–Mills’s flux is one global random variable; the edge entropy counts cut points, not information.
6. **Weak-coupling asymptotics with the heat-kernel constant** (derived; computed to machine precision). $S(t) = \frac{1}{2} \dim G \ln(1/t) + c_G + o(1)$ as $t \rightarrow 0^+$, with $c_G = \ln \text{vol}(G) - \frac{1}{2} \dim G \ln 2\pi + \frac{1}{2} \dim G$ in the Casimir-adapted metric, and $I(t) = \dim G/(2t^2)$, $C(t) \rightarrow \frac{1}{2} \dim G$; the e^{at} heat-kernel correction cancels identically in S . Closed forms for all three groups, verified to $\leq 2 \times 10^{-14}$ (Appendix A).
7. **Scope, stated as sharply as the results** (literature-verified). Everything here is $d = 2$ and purely the contested edge term; only the product $g^2 A$ is recoverable; and the $d = 4$ evidence runs *against* naive extrapolation (Secs. 5–6). Section 7 distills the model into a benchmark checklist for any future “coupling from entanglement” claim.

1.5 What is known and what is new

The reader should not mistake the inputs for the outputs. The formulas are known; the contribution is the estimation-theoretic reading and the explicit $SU(3)$ /capacity/ n -interval analysis. Table 2 draws the line item by item.

Known (literature, verified against sources)	New in this paper
Heat-kernel solution of 2d YM; all amplitudes as irrep sums [4, 5, 6, 7]	Exponential-family reading of $p_R(t)$: natural parameter $-t/2$, sufficient statistic $C_2(R)$, and everything that follows from it
Flux distribution $p_R(t) \propto d_R^2 e^{-(t/2)C_2}$ [9, eq. (pr)]	Identity chain packaged as theorems: $dS/dt = -(t/4) \text{Var}_t(C_2)$, $I(t) = \frac{1}{4} \text{Var}_t(C_2)$, $dS/dt = -t I(t)$ (Theorems 3, 5)
Edge entropy $S_n = H[p] + 2n \langle \ln d \rangle$ [9, 10, 11]; extended-Hilbert-space decomposition [8, 12, 13]	Capacity-of-entanglement identity $C(t) = t^2 I(t)$ in YM_2 (Prop. 6); weak-coupling constant c_G (App. A)
Qualitative n -interval point: extra intervals give no new information (abelian case, verbatim) [9]	n -cut derivative identity $dS_n/dt = -(t/4) \text{Var} - (n-1) \text{Cov}(\ln d, C_2)$ and the quantitative Fisher non-scaling $I_n = I_1$, with numerical demonstration (Theorem 7)
Capacity of entanglement as a concept [25, 26]; weak-coupling $\frac{1}{2} \dim G \log$ scaling [10, 20]	SU(3) computation; MLE/Cramér–Rao saturation demonstration; finite-sample bias and confidence-interval analysis (Sec. 4; Appendix B)

Table 2: Division of labor between the literature and this paper. The left column is literature-verified (sources and verbatim strings in `ym2/FORMULAS.md`); the right column is what this paper adds.

1.6 Relation to companion work

This paper is self-contained, but it grew out of two companion manuscripts, and one paragraph of context locates it; nothing below depends on them. Ref. [1] showed that the question “is a coupling hierarchy encoded in subsystem entanglement entropies?” is ill-posed for qubit subsystem entropies, which are gauge-frame dependent (row one of Table 1); Ref. [2] made the question well-posed with algebra-relative entropy (row two) and answered it negatively in a minimal fermionic toy, where all gauge-invariant entanglement is the classical edge/sector term. Both analyses terminate at the same object — the sector distribution at the boundary — as the only invariant carrier of coupling information in their settings. This paper analyzes that object in the one theory where it is exactly solvable. A further companion manuscript, since drafted [3], examines the $d = 4$ compact-U(1) descendant of the present mechanism (Sec. 6).

2 Setting and Verified Formulas

Status: literature-verified (sources and verbatim strings in `ym2/FORMULAS.md`).

2.1 The theory and the state

Pure Yang–Mills theory with compact connected Lie gauge group G in two Euclidean dimensions, in the standard heat-kernel formulation, is exactly solvable: the heat-kernel lattice action is self-reproducing under plaquette merging, so every amplitude on every surface has a closed form as a sum over irreps [4, 5, 6, 7]. On a closed surface Σ of area V and Euler characteristic χ ,

$$Z(\Sigma) = \sum_R (\dim R)^\chi \exp\left[-\frac{g^2 V}{2} C_2(R)\right],$$

with C_2 the quadratic Casimir. The theory has no propagating degrees of freedom; by invariance under area-preserving diffeomorphisms, g^2 and the areas enter only through products $g^2 \cdot (\text{area})$.

The state is the **Hartle–Hawking (sphere) state** on a spatial circle, the setting of Donnelly’s nonabelian analysis [9]: the wavefunction prepared by the path integral over a hemisphere (disk) of area $A/2$,

$$\psi(R) \propto (\dim R) e^{-(g^2 A/4) C_2(R)},$$

expressed in the character basis $|R\rangle$ of the physical Hilbert space (class functions of the circle holonomy). Everything depends on the single dimensionless parameter

$$t \equiv g^2 A, \quad A = \text{total sphere area} \quad (g^2 \text{ has dimension } 1/\text{area in } d=2).$$

2.2 The two formulas

Cutting the circle into a region A of n disjoint intervals ($2n$ cut points) and embedding into the extended Hilbert space (one copy of $L^2(G)$ per interval), the reduced density matrix is block-diagonal in the boundary irrep R (“electric flux”), with each block maximally mixed over the edge multiplicity space of dimension $(\dim R)^{2n}$ [9] (for the extended-Hilbert-space construction in general lattice gauge theories see also [8, 13]). The two formulas everything in this paper rests on are:

(F1) Flux distribution [9, eq. (pr)]:

$$p_R(t) = \frac{d_R^2 e^{-(t/2)C_2(R)}}{Z(t)}, \quad Z(t) = \sum_R d_R^2 e^{-(t/2)C_2(R)}, \quad (1)$$

where $d_R = \dim R$. This is the probability of measuring boundary flux R at the cut — in 2d YM, the Casimir of the boundary electric field is the *only* local gauge-invariant observable [9]. Note that p_R does not depend on n (Sec. 3.6).

(F2) Entanglement entropy, n intervals [9, eq. (YM2entropy)]; verbatim from the source:
 $S = -\sum_R p(R) (-\log p(R) + 2n \log \dim R)$:

$$S_n(t) = -\sum_R p_R \ln p_R + 2n \sum_R p_R \ln d_R = H[p(t)] + 2n \langle \ln d \rangle_t. \quad (2)$$

The same result follows from the Euclidean replica trick on the sphere with n cuts (Gromov–Santos [10, eq. (answ)], $\chi = 2$, $l = n$; the $l = 1$, $v = 0$ special case was obtained earlier by Velytsky [11]), up to the additive counterterm discussed below. We use $n = 1$ except in Sec. 3.6. In the three-term decomposition of [8], the first term of (2) is the classical (Shannon) piece, the second is the boundary log-dim piece, and the third term — distillable “nonlocal correlations” — **vanishes identically**: the entropy is entirely the edge/center term (see also the path-integral edge-mode treatment of [21]; the same “all edge, no bulk” structure appears in the finite-dimensional setting of [2]).

2.3 Group data and conventions

G	irreps R	$\dim R$	$C_2(R)$
compact U(1)	$n \in \mathbb{Z}$	1	n^2
SU(2)	$j \in \{0, \frac{1}{2}, 1, \dots\}$	$2j + 1$	$j(j + 1)$
SU(3)	$(p, q), p, q \in \mathbb{Z}_{\geq 0}$	$(p + 1)(q + 1)(p + q + 2)/2$	$(p^2 + q^2 + pq + 3p + 3q)/3$

SU(3) convention (verified against standard sources, June 2026 [33]). The Dynkin-label formulas above give $C_2(1, 0) = 4/3 = (N^2 - 1)/2N$ for the fundamental and $C_2(1, 1) = 3 = N$ for the adjoint — the normalization in which $\text{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$ in the fundamental. This is the same family as $C_2(j) = j(j + 1)$ for SU(2) (fundamental: $3/4 = (N^2 - 1)/2N$) and matches the convention of [9]. Both formulas are asserted at runtime in `ym2_su3.py` on the low-lying irreps ($\mathbf{1} = (0, 0)$, $\mathbf{3} = (1, 0)$, $\bar{\mathbf{3}} = (0, 1)$, $\mathbf{6} = (2, 0)$, $\mathbf{8} = (1, 1)$, $\mathbf{10} = (3, 0)$). Any rescaling $C_2 \rightarrow \kappa C_2$ — e.g., the 't Hooft-style convention of [10], which is $2\times$ ours for SU(2) — is the reparametrization $t \rightarrow \kappa t$ and changes nothing structural.

Additive ambiguity. S carries a t -independent additive ambiguity: the local counterterm $2lv$ of [10], the measure/bare-action constant footnoted in [9], the center-choice ambiguity of [12]. We fix it by the extended-Hilbert-space counting (trivial flux sector nondegenerate; $S \rightarrow 0$ as $t \rightarrow \infty$). The distribution $p_R(t)$ and all t -derivatives of S are ambiguity-free, and the reconstruction below uses only those.

3 Exact Results

3.1 Standing hypotheses and exponential-family structure

Status: derived (elementary); the observation that drives the paper.

All formal statements below share one set of hypotheses, which we state once:

Hypotheses (H). G is a *nontrivial* compact connected Lie group; the theory is pure 2d Yang–Mills with the heat-kernel Yang–Mills measure in the stated Casimir convention (Secs. 2, 2.3) on a sphere of area A , in the Hartle–Hawking state on the equatorial circle; $t = g^2 A \in (0, \infty)$; the region is $n \geq 1$ disjoint intervals; $S_n(t)$ is the extended-Hilbert-space entanglement entropy (2) with the additive counterterm fixed by the $v = 0$ convention of Sec. 2.3 (trivial flux sector nondegenerate, $S \rightarrow 0$ as $t \rightarrow \infty$); $p_R(t)$ is the flux distribution (1), which is convention-free.

Each qualifier earns its place. Nontriviality excludes the trivial group, whose single irrep gives $\text{Var}_t(C_2) = 0$: every identity below remains true but degenerates to $0 = 0$, and there is nothing to estimate. Connectedness guarantees infinitely many irreps with unbounded, non-constant Casimir, which is what the bijection and weak-coupling claims use; Remark 1 states what survives without it. The Casimir convention is part of the hypotheses, not a notational nicety; Remark 2 explains why, with torus factors the sharpest case.

Under (H), eq. (1) is a one-parameter exponential family in $\theta = -t/2$ with sufficient statistic $C_2(R)$ and base measure d_R^2 . Writing $\langle \cdot \rangle$, Var , Cov for moments under $p(t)$, the standard cumulant identities give

$$\frac{d\langle f(R) \rangle}{dt} = -\frac{1}{2} \text{Cov}_t(f, C_2) \quad \text{for any statistic } f, \quad \frac{d \ln Z}{dt} = -\frac{1}{2} \langle C_2 \rangle. \quad (3)$$

Everything in Secs. 3.2–3.6 is a corollary of (3).

Remark 1 (wider and narrower groups). The identities of Theorems 3–7 that involve only the exponential-family structure — dS_n/dt , $I(t)$, $C(t)$, and their relations — hold verbatim for *any* compact group, including finite and disconnected ones, whenever the Boltzmann weight is $e^{-(t/2)c(R)}$ for a fixed non-constant class statistic c (for compact disconnected Lie groups, C_2 itself; for finite groups, any chosen c , since no canonical Casimir exists). What fails without connectedness is the *range* statement: a finite group has finitely many irreps, so S is bounded, $S \not\rightarrow \infty$ as $t \rightarrow 0^+$,

and $t \mapsto S(t)$ is a bijection onto a bounded interval rather than onto $(0, \infty)$; the weak-coupling asymptotics of Appendix A, which use the heat-kernel divergence of the group manifold, have no analog. For a disconnected compact Lie group, the statements hold with $\dim G$ the dimension of the identity component and $\text{vol}(G)$ including the number of components. We therefore write “nontrivial compact connected Lie G ” in the theorems and treat these extensions as remarks, not as instances.

Remark 2 (torus factors and the Casimir convention). For groups with torus factors — $U(1)^k$ itself, or quotients such as $U(2) = (U(1) \times SU(2))/\mathbb{Z}_2$ — two points require care, which is why (H) names the Casimir convention explicitly. First, the irreps of a torus factor form a character *lattice* rather than a dominant-weight cone, and that lattice depends on the global form of G : for $U(2)$ the allowed (n, j) labels are correlated by the $\mathbb{Z}_2$ quotient, so the sum in (1) runs over the representation lattice of G itself, not over the product of the factors’ labels. Second, a torus direction has no canonical Casimir normalization: $C_2 = n^2$ for our compact $U(1)$ reflects a choice of charge unit, and rescaling it ($C_2 \rightarrow \kappa C_2$ on that factor alone) rescales the effective t of that factor independently of the others — for semisimple factors the same freedom exists per simple factor but is conventionally anchored to $\text{Tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$ (Sec. 2.3). Fixing the bi-invariant metric whose Laplace–Beltrami eigenvalues are the stated $C_2(R)$ fixes the heat-kernel Yang–Mills measure, and with it every constant in Appendix A. With that convention fixed, all statements hold verbatim for any nontrivial compact connected G .

3.2 Monotonicity: the convention-fixed S determines t

Status: derived; computed (residuals quoted).

From (1), the eigenvalues of the reduced density matrix in flux sector R ($n = 1$) are $\lambda_R = e^{-(t/2)C_2(R)}/Z$, each d_R^2 -fold degenerate, so $-\ln \lambda_R = \ln Z + (t/2)C_2(R)$ and

$$S(t) = \ln Z(t) + \frac{t}{2} \langle C_2 \rangle_t. \quad (4)$$

Differentiating with (3):

Theorem 3 (entropy–coupling monotonicity and inversion). *Assume (H) with $n = 1$. Then on $t \in (0, \infty)$:*

(a) (derivative identity)

$$\frac{dS}{dt} = -\frac{t}{4} \text{Var}_t(C_2) < 0, \quad (5)$$

where strict negativity holds because $p(t)$ has full support on infinitely many irreps with distinct Casimirs, so $\text{Var}_t(C_2) > 0$.

(b) (bijection) $Z(t)$ is analytic on $(0, \infty)$, so S is smooth and strictly decreasing, with $S \rightarrow \infty$ as $t \rightarrow 0^+$ (Appendix A) and $S \rightarrow 0$ as $t \rightarrow \infty$; hence $t \mapsto S(t)$ is a bijection $(0, \infty) \rightarrow (0, \infty)$.

(c) (what determines t) The convention-fixed extended-Hilbert-space entropy — or equivalently the ambiguity-free flux distribution $p_R(t)$ — determines t exactly.

Proof. (a) Differentiate (4) using (3): $dS/dt = -\frac{1}{2} \langle C_2 \rangle + \frac{1}{2} \langle C_2 \rangle + \frac{t}{2} (-\frac{1}{2} \text{Var}_t(C_2)) = -\frac{t}{4} \text{Var}_t(C_2)$. (b) Analyticity of Z follows from uniform convergence of the irrep sum on compact subsets of $(0, \infty)$ (Casimir growth dominates dimension growth); the $t \rightarrow 0^+$ divergence is the heat-kernel asymptotics of Appendix A; as $t \rightarrow \infty$ the trivial sector dominates and $S \rightarrow 0$ in the $v = 0$ convention. (c) is immediate from (b) given the convention, and from injectivity of $t \mapsto p(t)$ (e.g. via $\langle C_2 \rangle_t$ strictly decreasing) without it. $\square$

Remark 4 (the convention dependence is essential, and quarantined). Part (c) is stated with care because S carries a t -independent additive counterterm ambiguity (Sec. 2.3). *Without* fixing the convention, the exact entropy determines t only up to that ambiguity — an observer handed “the value of S ” in an unknown convention learns t only after one calibration point. The objects that determine t with no convention at all are the flux distribution $p_R(t)$, any t -derivative of S , and finite flux samples (Sec. 4). Every reconstruction channel used in this paper is of that ambiguity-free kind except the explicit entropy-inversion demonstration of Sec. 4.2, which declares its convention.

Computed verification. On $t \in [0.05, 20]$ (400-point geometric grid): S strictly decreasing for $U(1)$, $SU(2)$, $SU(3)$; the analytic derivative (5) matches finite differences to better than 0.3% ($SU(2)$, $U(1)$) and 0.6% ($SU(3)$) everywhere on the grid (grid-stencil error), and to 2.4×10^{-10} with dedicated central differences (Sec. 3.6); max dS/dt on the grid: -6.2×10^{-3} ($SU(2)$), -4.5×10^{-4} ($U(1)$), -2.6×10^{-4} ($SU(3)$) — bounded away from zero. Practical inversion (given exact S , root-find t) recovers t to $\leq 9 \times 10^{-14}$ at six test points per group. The Shannon-only piece $H(t) = S - 2\langle \ln d \rangle$ is *not* manifestly monotonic ($dH/dt = -(t/4) \text{Var}(C_2) + \text{Cov}(\ln d, C_2)$, a competition); numerically it is also strictly decreasing for all three groups on the grid, but only the full S has the analytic proof. We flag this honestly because H , not S , is the piece [9] identifies with uncertainty in gauge-invariant observables.

The weak-coupling behavior is universal, and it is what justifies $S \rightarrow \infty$ in Theorem 3: $Z(t) = \sum_R d_R^2 e^{-(t/2)C_2(R)}$ is the heat kernel of the group manifold G evaluated at the identity, whose short-time behavior is the standard flat divergence $Z(t) \simeq \text{vol}(G) (2\pi t)^{-\dim G/2}$ [32, 7], so $S(t) = \frac{1}{2} \dim G \ln(1/t) + c_G + o(1)$ as $t \rightarrow 0$. Appendix A derives the constant, $c_G = \ln \text{vol}(G) - \frac{1}{2} \dim G \ln 2\pi + \frac{1}{2} \dim G$, evaluates it in closed form for all three groups, and verifies it numerically to $\leq 2 \times 10^{-14}$; the same expansion gives $I(t) = \dim G/(2t^2)$ and $C(t) \rightarrow \frac{1}{2} \dim G$. This is consistent with the weak-coupling scaling of [10], and formally the same $\frac{1}{2} \dim G \log(\text{coupling})$ structure as Radićević’s $d = 3$ universal term [20].

3.3 Fisher information: reconstructability = susceptibility

Status: derived; the framing we believe is new (novelty scan recorded in the repository notes).

Theorem 5 (Fisher-information identity). *Assume (H). The Fisher information about t of a single flux measurement at one cut is*

$$I(t) = \mathbb{E}[(\partial_t \ln p_R)^2] = \frac{1}{4} \text{Var}_t(C_2), \quad (6)$$

so the Cramér–Rao bound for any unbiased estimator of t from N i.i.d. flux samples is $\text{Var}(\hat{t}) \geq 1/(NI(t)) = 4/(N \text{Var}_t(C_2))$; and, comparing with (5),

$$\boxed{\frac{dS}{dt} = -t I(t).} \quad (7)$$

Proof. For the family (1), $\partial_t \ln p_R = \frac{1}{2}(\langle C_2 \rangle - C_2(R))$, whose square has expectation $\frac{1}{4} \text{Var}_t(C_2)$; (7) is (5) divided by (6). $\square$

The entropy’s sensitivity to the coupling and the statistical information the flux carries about the coupling are the same function of t , up to the factor $-t$. Physically, $\text{Var}_t(C_2)$ is the flux susceptibility of the vacuum — the defensible core of any coupling-from-entanglement claim. Eq. (7) says that in $d = 2$ the identification is exact: *entanglement susceptibility is Fisher information*.

The flux measurement is quantum-optimal. A referee may ask whether some other measurement on the interval extracts more information about t . No: the reduced density matrices $\{\rho_A(t)\}$ are all diagonal in the same flux-block basis (eq. (1) with t -independent eigenvectors), so the family is mutually commuting, and the quantum Fisher information of $\rho_A(t)$ equals the classical Fisher information of its eigenvalue distribution [31]. Within a block the state is maximally mixed with t -independent weights $1/d_R^2$, carrying no information; the eigenvalue distribution over blocks is $p_R(t)$. Hence $\text{QFI}(t) = I(t) = \frac{1}{4} \text{Var}_t(C_2)$: the flux measurement saturates the quantum Cramér–Rao bound among all measurements on the interval’s algebra, and (6) is not an artifact of a convenient measurement choice.

3.4 Two reconstruction channels, one identity

Inverting an entropy and sampling a flux are operationally different acts, and the paper’s headline claims should be read against that distinction. There are two channels:

- (A) **Inversion from the exact entropy.** Given the exact value of the *convention-fixed* $S(t^*)$ — one real number, no samples — Theorem 3(b) recovers t^* by root finding. This channel *requires* the additive-counterterm convention (Remark 4); handed S in an unknown convention, it returns t only up to calibration. It is also idealized: no experiment returns an exact entropy; entropy is not an observable but a functional of the state, estimated in practice through many measurements.
- (B) **Estimation from finite flux samples.** An observer who measures the boundary flux N times draws $R_1, \dots, R_N \sim p_R(t^*)$ and forms the MLE $\hat{t}$ (Sec. 4). This channel is convention-free — $p_R(t)$ carries no counterterm — and operational: the flux is a measurable gauge-invariant observable, indeed the only local one here [9].

These are different protocols with different presuppositions, and conflating them is the error the review literature warns against (“measuring entanglement entropy” usually means channel (B)-style statistics dressed in channel (A) language). What ties them together is Theorem 5: the sensitivity of channel (A) — how fast S moves when t moves, $|dS/dt| = tI(t)$ — and the information rate of channel (B) — $I(t)$ per sample — are the same function. Concretely, an uncertainty δS in the entropy propagates to $\delta t = \delta S/(tI(t))$, while N flux samples give $\delta t = 1/\sqrt{NI(t)}$: both channels are governed by the single function $I(t) = \frac{1}{4} \text{Var}_t(C_2)$, which is why we report them side by side without identifying them.

3.5 Capacity of entanglement

Status: derived; computed. Concept from [25, 26]; we found no prior YM_2 appearance.

The capacity of entanglement is the variance of the modular Hamiltonian, $C(t) \equiv \text{Var}(-\ln \rho_A)$ — the entanglement analog of heat capacity [25, 26].

Proposition 6 (capacity identity). *Assume (H) with $n = 1$. Then*

$$C(t) = \frac{t^2}{4} \text{Var}_t(C_2) = t^2 I(t), \quad \frac{dS}{dt} = -\frac{C(t)}{t}, \quad \text{and} \quad C(t) \rightarrow \frac{1}{2} \dim G \text{ as } t \rightarrow 0^+. \quad (8)$$

Proof. $-\ln \lambda_R = \ln Z + (t/2)C_2(R)$ is affine in C_2 , so $\text{Var}(-\ln \rho_A) = (t^2/4) \text{Var}_t(C_2) = t^2 I(t)$; the second relation is (7); the limit is Appendix A, eq. (12). $\square$

In this model the capacity of entanglement *is* the Fisher information in disguise — a clean instance of the thermodynamic analogy (capacity $\leftrightarrow$ heat capacity $\leftrightarrow$ fluctuation $\leftrightarrow$ information) becoming an identity. Computed: $C(t) \rightarrow \frac{1}{2} \dim G$ as $t \rightarrow 0^+$, equal to 0.5000, 1.5000, 4.0000 (U(1), SU(2), SU(3)) at $t = 0.05$ to the precision quoted in the results files; equivalently $S \simeq \frac{1}{2} \dim G \ln(1/t)$ at weak coupling, with the capacity plateau ending around $t \sim 1$ (SU(3) still 3.995 at $t \approx 3.9$).

Prior work. Capacity of entanglement was introduced in [25], studied systematically in QFT and holography in [26], and computed for local-operator excitations of 4d free Yang–Mills in [27]. We also note the contact with the symmetry-resolved entanglement literature: the flux-block decomposition of ρ_A is precisely a charge/sector-resolved structure in the sense of [28, 29], and the capacity of entanglement has been used as a probe of RG flows and symmetry resolution in [30]; our eq. (8) is the YM₂ instance in which the sector distribution alone carries all of the t -dependence. Searches (June 2026; queries recorded in the repository notes) found no prior computation in 2d Yang–Mills and no prior statement of (6)–(8) there; we flag that the entanglement entropy of YM₂ itself is thoroughly studied [10, 9, 22, 23], so (8) is best read as low-hanging fruit made visible by the Fisher framing, not as a difficult result.

3.6 n intervals: the entropy scales, the information does not

Status: formula (2) literature-verified; identity (9) derived; non-scaling of information derived and computed; the qualitative point is already in [9].

Setting adopted (stated carefully, since the cut topology matters). The spatial manifold is the circle in the HH sphere state; region $A = n$ disjoint intervals, complement = n intervals, $2n$ cut points. The physical state is a class function of the single circle holonomy, $\psi = \sum_R c_R |R\rangle$; iterating Donnelly’s embedding map, the character state $|R\rangle$ embeds into the $2n$ -interval extended Hilbert space with one d_R -dimensional index pair per cut point, all carrying the *same* R . The reduced density matrix is therefore block-diagonal in **one global flux label** R with the n -independent distribution $p_R(t)$ of eq. (1), each block maximally mixed of dimension d_R^{2n} . This yields (2), whose log-dim term “appears once for each of the $2n$ points on the boundary of A ” [9, verbatim]. The replica computation of [10] on the sphere with $l = n$ cuts gives the same t -dependence. (On higher-genus surfaces or for non-HH states the sector structure is richer [23]; we do not use those settings.)

Theorem 7 (n -cut identity and information non-scaling). *Assume (H) with $n \geq 1$ intervals. Then:*

(a) (entropy derivative)

$$\frac{dS_n}{dt} = -\frac{t}{4} \text{Var}_t(C_2) - (n-1) \text{Cov}_t(\ln d, C_2). \quad (9)$$

(b) (information non-scaling) *The Fisher information about t of the joint flux measurement at all $2n$ cut points equals that of one cut: $I_n(t) = I_1(t) = \frac{1}{4} \text{Var}_t(C_2)$ for every n .*

Proof. (a) From $S_n = S_1 + 2(n-1)\langle \ln d \rangle$ (eq. (2)) and the cumulant identity (3) applied to $f = \ln d$, together with (5). (b) The $2n$ cut-point fluxes of one prepared circle are copies of the single global R (the reduced state is block-diagonal in one shared label, as recalled above): the joint distribution is supported on the diagonal $\{(R, \dots, R)\}$ with weights $p_R(t)$, so the joint likelihood equals the single-cut likelihood and the Fisher informations coincide. $\square$

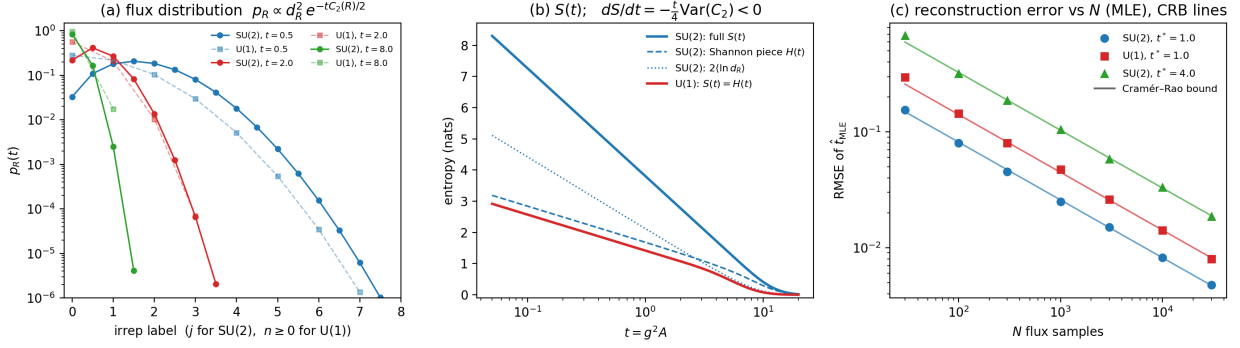


Figure 1: U(1) and SU(2) (script `ym2_flux.py`, seed 20260609). (a) Flux distribution $p_R(t)$ of eq. (1) at $t \in \{0.5, 2, 8\}$. (b) Entropy $S(t)$, its Shannon piece $H(t)$, and the edge term: all strictly decreasing, Theorem 3. (c) RMSE of the maximum-likelihood estimate $\hat{t}$ versus number of flux samples N , against the Cramér–Rao lines $\sqrt{1/(NI(t^*))}$ (Appendix B).

Computed. Identity (9) matches central finite differences to $\leq 2.4 \times 10^{-10}$ relative over 48 (group, t , n) combinations (U(1), SU(2), SU(3); $t \in \{0.3, 1, 3, 8\}$; $n \in \{1, 2, 3, 4\}$). $\text{Cov}_t(\ln d, C_2) > 0$ on the full 400-point grid for SU(2) and SU(3) (minima 1.1×10^{-3} and 4.3×10^{-5} respectively) and is identically 0 for U(1) ($d \equiv 1$); so S_n is strictly decreasing in t for every n in all cases computed — each additional pair of cut points makes the entropy *more* coupling-sensitive, by $(n-1) \text{Cov}$. Donnelly states the qualitative information point verbatim for the abelian case [9]: “the entropy is independent of the number of intervals traced out... Having access to an additional interval therefore does not change the amount of information one can acquire about the state.” (In the nonabelian case the *entropy* does grow with n — but only through the $2n \langle \ln d \rangle$ edge term, which [9] notes “is not associated to uncertainty in gauge-invariant observables.”)

Computed demonstration. $M = 400$ trials, each preparing $N = 1000$ independent circles cut into $n = 3$ intervals (6 cut points, 6000 flux readings, 1000 independent values). The MLE from the 6000-reading dataset is *identical* to the MLE from the 1000 unique samples (duplication leaves the sufficient statistic unchanged; verified to 10^{-14} relative, exact as estimators), and its standard deviation matches the N -preparation Cramér–Rao bound, not the hypothetical $2nN$ -sample bound: $\text{sd}/\text{CRB}_N = 0.93$ (SU(3)), 0.97 (SU(2)); $\text{sd}/\text{CRB}_{2nN} = 2.27, 2.37 \approx \sqrt{6} \cdot (\text{sd}/\text{CRB}_N)$.

The moral is worth the sentence: **in 2d Yang–Mills, entropy counts cut points; information counts independent preparations.** The edge term $2n \langle \ln d \rangle$ — the dominant part of S_n for large n — carries no additional information about the coupling beyond what one cut already provides. A referee who suspects that “more entanglement entropy” must mean “better reconstruction” has it exactly backwards here, and the structure of the reduced state says why: 2d YM’s flux is one global random variable. (Compare the flux-tube entanglement of [24], which likewise depends on the number of boundary crossings and $\dim R$ but not on lengths.)

4 Statistical Reconstruction at the Cramér–Rao Limit

Status: derived; finite- N behavior computed (deterministic seed 20260609; protocol details, tables, and scripts in Appendix B).

The operational protocol is channel (B) of Sec. 3.4: an observer who can measure only the gauge-invariant flux observable at one cut draws N i.i.d. samples $R_1, \dots, R_N$ from $p_R(t^*)$ (N independent

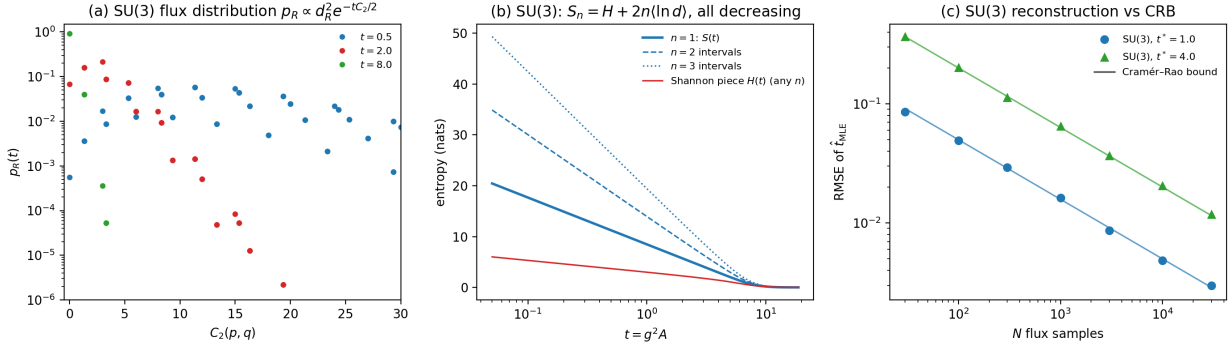


Figure 2: SU(3) and the n -interval analysis (script `ym2_su3.py`, seed 20260609). (a) SU(3) flux distribution over Casimir values at $t \in \{0.5, 2, 8\}$. (b) $S_n = H + 2n(\ln d)$ for $n = 1, 2, 3$ intervals and the n -independent Shannon piece $H(t)$: the entropy scales with n , the information does not (Sec. 3.6). (c) SU(3) reconstruction error versus N against the Cramér–Rao bound (Appendix B).

preparations; Sec. 3.6 is why repeated cuts of one preparation do not count) and estimates t . Because (1) is an exponential family, the maximum-likelihood estimator is moment matching — solve $\langle C_2 \rangle_t = N^{-1} \sum_i C_2(R_i)$, which has a unique root since $\langle C_2 \rangle_t$ is strictly decreasing by (3) — and standard theory guarantees consistency, asymptotic normality, and Cramér–Rao saturation as $N \rightarrow \infty$. None of this needs simulation: it is the expected behavior of a regular one-parameter exponential family, and we present it as such.

What asymptotic theory does not supply is the finite- N picture, and that is what the Monte Carlo experiments of Appendix B verify: root-mean-square error on the Cramér–Rao line within Monte-Carlo resolution from $N = 10^2$ to 3×10^4 for U(1), SU(2), and SU(3) — e.g. t recovered to 1.6% from 10^3 samples and 0.48% from 10^4 at $t^* = 1$ for SU(3) — and Wald and profile-likelihood confidence intervals at nominal coverage in all 27 (group, t^* , N) cells. The one genuinely finite-sample item is the bias: expanding the moment equation around t^* with the cumulant identities of the family gives the first-order law

$$\mathbb{E}[\hat{t}] - t^* = b_1(t^*) + O(N^{-2}), \quad b_1(t^*) = \frac{\kappa_3(C_2)}{N \text{Var}_{t^*}(C_2)^2}, \quad (10)$$

with κ_3 the third central moment of C_2 under $p(t^*)$, and the measured bias matches (10) within Monte-Carlo error in 26 of 27 cells (the exception is strong coupling at $N = 10^2$, where higher-order terms are still comparable; Appendix B). Bias is negligible against the statistical error everywhere ($|b_1| \leq 0.15 \sigma_{CR}$ at $N = 10^2$, falling as $N^{-1/2}$ relative), so the reconstruction error is variance-dominated and the Cramér–Rao comparison is the right one.

4.1 SU(3), and why bigger groups are easier

SU(3) is included deliberately, and the statement it supports is confined to two dimensions: **in the $d = 2$ world, the dimensionless coupling $t = g^2 A$ of an SU(3) gauge theory is reconstructible from gauge-invariant edge-sector statistics at the Cramér–Rao limit.** No claim is made about four-dimensional QCD; Section 5 confines the sentence with full force.

The group-size trend is itself informative. At weak coupling $I(t) \approx \dim G / (2t^2)$ (from $C(t) \rightarrow \frac{1}{2} \dim G$ and (8)): $I(1) = 0.5, 1.5, 4.0$ for U(1), SU(2), SU(3). Each flux sample from a larger group carries more information about the coupling, because the Casimir spectrum is richer and the d_R^2

degeneracy weighting spreads p_R over more, better-separated sectors: in the one solvable setting, the *non-abelian* couplings are the easy ones.

4.2 Inversion from the exact entropy (channel (A))

Independent of sampling, and idealized: given the exact value of $S(t^*)$ alone (one number, no samples), root-finding on the strictly monotonic $S(\cdot)$ recovers t^* to $\leq 9 \times 10^{-14}$ at all test points $t^* \in \{0.3, 0.7, 1.5, 3, 6, 12\}$ for all three groups — the bijection of Theorem 3(b) in practice. We emphasize that this is the idealized channel (A) of Sec. 3.4 — no experiment returns an exact entropy — and that it uses the $v = 0$ normalization (Sec. 2.3); an observer uncertain about the additive counterterm should use derivative or sampling data, which are ambiguity-free (Remark 4).

5 Scope: What This Does and Does Not Show

Status: the deflationary half of the thesis; literature-verified where cited.

1. **It is $d = 2$, and the solvability is the unrepresentativeness.** There are no propagating gluons; the entire entropy is the edge/center term (the distillable term of [8] vanishes identically). The features that make the theorem exact — one global flux variable, no transverse physics, area-only dynamics — are precisely the features absent in $d = 4$. Nothing here is evidence that $d = 4$ couplings live in entanglement; it is the exact statement of what the best case looks like.
2. **The carrier of the signal is contested as “entanglement.”** The edge term is non-distillable and drops out of mutual information and relative entropy in the continuum limit [14, 15]; it depends on the choice of center/boundary algebra [12]; and although the edge-mode reading has independent support — it matches the contact term of Kabat [19] in the Maxwell case [16, 17] — Casini–Huerta–Magán–Pontello [18] argue that in $d = 4$ Maxwell theory the analogous edge assignment is regularization-dependent, with the physical (anomaly-restoring) effect coming from charged vacuum fluctuations — an effect *independent of the coupling’s value*. Our reconstruction reads information out of exactly the term whose physical status is the live dispute. Two mitigations, honestly weighed: (i) the flux distribution $p_R(t)$ itself is an unambiguous, gauge-invariant object whatever one calls its Shannon entropy — the statistical reconstruction of Sec. 4 needs only p_R , not an entropy interpretation; (ii) but an operationalist who insists on distillable entanglement as the only physical entanglement will say this paper reconstructs the coupling from *superselection-sector statistics*, not from entanglement. We think that reading is defensible — it is why the title says “edge-sector statistics” rather than “entanglement,” and any entanglement gloss on the results should be heard with that asterisk.
3. **Only $g^2 A$ is recoverable — reflection has a gauge orbit of its own.** By area-preserving diffeomorphism invariance, g^2 and A enter all observables only through $t = g^2 A$ [10]; “the coupling” is recoverable only relative to a fixed fiducial area. This is the $d = 2$ echo of the $d = 4$ obstruction that for noncompact free Maxwell, g can be removed entirely by field redefinition: couplings are scheme-relative quantities, and entanglement data can at best pin down their invariant combinations.
4. **Reflection, not determination — the precise statement.** What is proven: the map $t \mapsto p(t)$ is injective (Theorem 3) and its inversion from samples is statistically efficient (Sec. 4). What is presupposed: the group G , the Hartle–Hawking preparation, and the

functional form of the family (1) — i.e., the entire theory up to one number. An inference of this shape can never show that couplings *are* entanglement properties; it shows that a known theory’s coupling leaves a complete, efficiently readable fingerprint in invariant data. Determination would require, at minimum, an induced-coupling framework in which $1/g^2$ is *defined* by an entanglement functional (the gauge analog of Sakharov/Jacobson induced gravity) — no such framework exists in the literature we verified.

5. **The additive ambiguity is real but quarantined.** $S(t)$ carries the t -independent counterterm ambiguity (Sec. 2.3). All reconstruction channels used here — $p_R(t)$, t -derivatives of S , sampled fluxes — are ambiguity-free. A claim that needed the absolute value of S would not be.

6 What $d = 3$ and $d = 4$ Would Require

Status: literature-verified program statement; nothing here is solvable by the methods of this paper.

$d = 3$. The entropy is no longer all edge: the distillable term of [8] is nonzero (a propagating photon/gluon), and the boundary flux becomes a *field* on the entangling curve rather than one label. Radićević’s weak-coupling analysis [20] shows the coupling still enters a universal term, $\frac{1}{2} \dim G \cdot \log(e^2 r)$ for a region of size r — formally the same $\frac{1}{2} \dim G \cdot \log(\text{coupling})$ structure as our weak-coupling $S(t)$. A $d = 3$ reconstruction program would need: (i) the joint distribution of boundary fluxes (at weak coupling a Gaussian field with covariance $\langle E_\perp E_\perp \rangle$ set by e^2) — note that unlike our single global R , *this* distribution genuinely has more independent components for larger boundaries, so the n -cuts no-go of Sec. 3.6 is a $d = 2$ peculiarity, not a general law; (ii) separation of the universal log coefficient from non-universal perimeter pieces; (iii) a stance on the center choice, which in $d \geq 3$ affects the sufficient statistics themselves [12].

$d = 4$. We keep this short and programmatic, because three verified facts already frame it. (i) For noncompact free Maxwell, vacuum entanglement cannot depend on g at all (field redefinition). (ii) For compact $U(1)$, the coupling enters through flux-sector (zero-mode) weights — a finite, topological term [17] — and through compactness effects on the edge partition function. (iii) With charged matter, the one sharp computation [18] finds the log-coefficient shift universal — independent of the coupling’s value and of the charge masses. The live possibilities are therefore: the coupling enters only through compactness/topology, or only through scale-dependence of coarse-grained flux statistics, or not at all in any distillable quantity.

The one calculation this points to, and its status. The direct $d = 4$ descendant of Theorem 5 is: **compute the Fisher information about e^2 carried by the compact- $U(1)$ edge-flux sector distribution across a closed entangling surface in $d = 4$** (e.g. a sphere, with the zero-mode/flux-sector weights of [17]). The sector weights are an exponential family in $1/e^2$ with sufficient statistic the flux quantum number squared, so $I(e^2)$ is computable in closed form once the weights are fixed; the questions the calculation decides are whether $I(e^2) \neq 0$ survives the continuum/large-surface limit, which regularization ambiguities it inherits (item 6 of the checklist below), and how it scales with the entangling-surface geometry (item 7). That computation has since been carried out and is reported in a drafted companion manuscript [3]; its character, in one sentence and without importing detailed results: the exact $d = 2$ structure has no global $d = 4$ analog, but phase-by-phase *asymptotic* exponential families survive, with a finite topological zero-mode Fisher information — a weak, topological descendant of the $d = 2$ mechanism, consistent with the deflationary reading of this paper. The present paper stands independently of [3]: nothing

here uses its results, and the $d = 2$ identity chain remains the exact benchmark such computations are measured against.

7 2d Yang–Mills as a Benchmark: a Checklist

The $d = 2$ result fixes the gold standard against which any future “coupling from entanglement” claim — in $d = 3$, $d = 4$, holography, or analog experiments — should be measured. In this model every item below has an exact answer; a future claim should specify all seven, and the burden of stating them falls on the claimant:

1. **Observable algebra.** Which gauge-invariant algebra (and which center choice [12]) is the entanglement relative to? [Here: class functions of the holonomy; trivial choice ambiguity, flagged.]
2. **Sector distribution.** What is the analog of $p_R(t)$ — the probability distribution over superselection sectors at the cut — and is it convention-free? [Here: eq. (1); yes.]
3. **Sufficient statistic.** Through which statistic does the coupling enter the sector distribution? [Here: $C_2(R)$, exponential family in $-t/2$.]
4. **Fisher information.** What is $I(\text{coupling})$ for the best gauge-invariant measurement, and does it saturate the quantum Fisher information? [Here: $I(t) = \frac{1}{4} \text{Var}_t(C_2) = \text{QFI}$.]
5. **Reconstruction map.** Is the claim inversion of an exact functional (channel (A)) or estimation from samples (channel (B)) — and at what efficiency? [Here: both, tied by (7); Cramér–Rao saturated.]
6. **Ambiguity/counterterm dependence.** Which parts of the claim survive the additive counterterm and regularization ambiguities of gauge-theory entanglement? [Here: everything except the absolute value of S ; Remark 4.]
7. **Sample-vs-cut scaling.** Does more entangling boundary mean more information, or just more entropy? [Here: more cuts give exactly nothing, Theorem 7 — entropy counts cut points, information counts independent preparations.]

The checklist is deliberately deflationary: items 1, 2, and 6 are where frame-dependent or scheme-dependent claims die (cf. [1, 2]), and item 7 is where “more entanglement” rhetoric dies. A $d = 4$ proposal that cannot fill in the seven lines is not yet a claim.

8 Conclusion

In two-dimensional Yang–Mills on a sphere, the edge-sector distribution at an entangling cut is a one-parameter exponential family in the coupling, and everything this paper proves is a corollary of that single fact. The coupling is reflected *completely* (the map $t \mapsto p(t)$ is injective, and the convention-fixed entropy is a bijection of $(0, \infty)$ onto itself), *identifiably* (one global flux variable, with no information inflation from extra cuts), and *efficiently* (Cramér–Rao saturation, for $U(1)$, $SU(2)$, and $SU(3)$, with larger groups easier). The exact identities

$$\frac{dS}{dt} = -\frac{t}{4} \text{Var}_t(C_2) = -t I(t) = -\frac{C(t)}{t}$$

say it in one line: in $d = 2$, entanglement susceptibility, flux susceptibility, Fisher information, and capacity of entanglement are one function, and that function knows the coupling.

What none of this shows is determination. The inference presupposes the theory; the recovered number is the scheme-invariant product $g^2 A$; the carrier is a term whose entanglement credentials are contested — the defensible reading is superselection-sector statistics; and the $d = 4$ evidence is, if anything, adverse. The honest summary is therefore a conditional: *if* gauge couplings are encoded in vacuum entanglement in higher dimensions, the encoding must be of the kind exhibited here — gauge-invariant sector statistics, not frame-dependent subsystem entropies — and whether any $d = 4$ version of that encoding exists beyond topological zero modes is the open question, taken up for compact $U(1)$ in [3]. Section 7 is the checklist any successor claim must fill in.

A Weak-Coupling Asymptotics

Status: derived; computed to machine precision (script `paper3/weak_coupling_check.py`).

Proposition 8 (weak-coupling expansion). *Assume (H) with $n = 1$. As $t \rightarrow 0^+$,*

$$S(t) = \frac{1}{2} \dim G \ln \frac{1}{t} + c_G + \varepsilon_S(t), \quad c_G = \ln \text{vol}(G) - \frac{1}{2} \dim G \ln 2\pi + \frac{1}{2} \dim G, \quad (11)$$

$$I(t) = \frac{\dim G}{2t^2} (1 + \varepsilon_I(t)), \quad C(t) = t^2 I(t) \rightarrow \frac{1}{2} \dim G, \quad (12)$$

where $\text{vol}(G)$ is the Riemannian volume of G in the bi-invariant metric whose Laplace–Beltrami eigenvalue on irrep R is the $C_2(R)$ used throughout, and the remainders $\varepsilon_S, \varepsilon_I$ are exponentially small in $1/t$ (not power-law).

Proof. By Peter–Weyl, $Z(t) = \sum_R d_R^2 e^{-(t/2)C_2(R)} = \text{vol}(G) K_{t/2}(e)$, with K_s the heat kernel of the Laplace–Beltrami operator on the group manifold and $K_s(e)$ its value at the identity. Its short-time behavior is $K_s(e) = (4\pi s)^{-\dim G/2} e^{bs} (1 + \text{exp. small})$, with b the curvature constant ($b = |\rho|^2$ in the standard normalization; see [32] for the general homogeneous-space expansion), so $Z(t) = \text{vol}(G) (2\pi t)^{-\dim G/2} e^{bt/2} (1 + \text{exp. small})$.

The key structural fact is a cancellation: *if $Z(t) = c_0 t^{-d/2} e^{\beta t}$ exactly, then S, I, C are independent of β :*

$$\langle C_2 \rangle = -2 \frac{d \ln Z}{dt} = \frac{d}{t} - 2\beta, \quad S = \ln Z + \frac{t}{2} \langle C_2 \rangle = \ln c_0 + \frac{d}{2} - \frac{d}{2} \ln t,$$

$$\text{Var}_t(C_2) = -2 \frac{d \langle C_2 \rangle}{dt} = \frac{2d}{t^2}, \quad I(t) = \frac{d}{2t^2}, \quad C(t) = t^2 I(t) = \frac{d}{2},$$

all exactly. The linear-in- t exponent of the heat kernel therefore drops out of every quantity in the proposition, and the only corrections are the exponentially small ones, which survive differentiation as exponentially small terms. Substituting $c_0 = \text{vol}(G)(2\pi)^{-\dim G/2}$ gives c_G . $\square$

Closed-form constants. The metric normalization is fixed by the Casimir convention of Sec. 2.3:

- $U(1)$: irreps $n \in \mathbb{Z}$, $C_2 = n^2$; the circle of circumference 2π has Laplacian spectrum n^2 , so $\text{vol} = 2\pi$ and $c_{U(1)} = \frac{1}{2} + \frac{1}{2} \ln 2\pi = 1.41893853 \dots$
- $SU(2)$: $C_2(j) = j(j+1)$; the 3-sphere of radius 2 has spectrum $l(l+2)/4 = j(j+1)$ with $l = 2j$, so $\text{vol} = 2\pi^2 \cdot 2^3 = 16\pi^2$ and $c_{SU(2)} = \ln 16\pi^2 - \frac{3}{2} \ln 2\pi + \frac{3}{2} = 3.80523289 \dots$

- SU(3): in this normalization $\text{vol}(\text{SU}(3)) = 8\sqrt{3}(2\pi)^5$, consistent with the standard invariant-volume formulas and verified numerically below; hence $c_{\text{SU}(3)} = \ln(8\sqrt{3}) + \ln 2\pi + 4 = 8.46662475\dots$

Computed verification (`weak_coupling_check.json`; grid $t \in [10^{-3}, 0.2]$, all three groups): $|S(t) + \frac{1}{2} \dim G \ln t - c_G| \leq 2 \times 10^{-14}$ and $|2t^2 I(t) / \dim G - 1| \leq 4 \times 10^{-15}$ at every grid point — machine precision, with no visible t -dependence. The corrections are indeed exponentially small rather than power-law, exactly as the cancellation argument predicts: at $t = 0.2$ a power-law correction would be conspicuous, and there is none. (This sharpens the 10^{-6} -level witness quoted in earlier drafts, which was limited by the verification grid, not by the physics.) For U(1) the statement is elementary: $Z(t) = \theta_3(0, e^{-t/2})$, and Poisson resummation gives $Z = \sqrt{2\pi/t} (1 + 2e^{-2\pi^2/t} + \dots)$, with $e^{-2\pi^2/t} < 10^{-42}$ already at $t = 0.2$.

B Monte Carlo Verification at Finite N

Status: computed (deterministic seed 20260609; scripts listed in Appendix C).

Cramér–Rao saturation as $N \rightarrow \infty$ is guaranteed for this exponential family (Sec. 4); this appendix records the finite- N verification. The protocol is that of Sec. 4: each batch draws N i.i.d. flux samples from $p_R(t^*)$ and forms the MLE $\hat{t}$ by moment matching; M independent batches per cell give the RMSE, bias, and coverage estimates below.

B.1 RMSE against the Cramér–Rao bound

At $t^* = 1$ ($M = 400$ batches per cell; RMSE of $\hat{t}$; CRB = $\sqrt{1/(NI(t^*))}$; ratio = RMSE/CRB):

group	$I(t^*=1)$	$N = 10^2$	$N = 10^3$	$N = 10^4$	ratio at $10^3 / 10^4$
U(1)	0.500	0.1437	0.0472	0.0141	1.05 / 1.00
SU(2)	1.500	0.0800	0.0250	0.0082	0.97 / 1.00
SU(3)	4.000	0.0490	0.0162	0.0048	1.03 / 0.97

Errors fall as $N^{-1/2}$ along the CR line (Figs. 1c, 2c). Across all $t^* \in \{0.5, 1, 2, 4\}$ at $N = 3 \times 10^4$: $\text{Var}(\hat{t})/\text{CRB} \in [0.93, 1.08]$ (SU(2), U(1), 8 cells) and $[0.89, 1.07]$ (SU(3), 4 cells) — saturation within Monte-Carlo resolution ($\pm \sim 7\%$ at 400 batches). Bias is $O(1/N)$ and negligible ($|\text{bias}| \leq 1.4 \times 10^{-4}$ at $N = 3 \times 10^4$, SU(3), $t^* = 1$). SU(3) reconstruction at $t^* \in \{0.5, 1, 2, 4\}$ and $N = 10^3$ achieves RMSE $\{0.0080, 0.0162, 0.0318, 0.0650\}$, i.e. 1.6% relative at $t^* = 1$ and 10^3 samples, 0.48% at 10^4 . No clamped batches occurred for SU(3) (the d_R^2 degeneracy pressure makes all-trivial-sector samples vanishingly rare); the single pathological batch in the whole study remains the U(1), $t^* = 4$, $N = 30$ cell documented in the results files.

B.2 Finite-sample bias and confidence intervals

Script `paper3/mle_bias_ci.py`, seed 20260609; $M = 2000$ batches per cell, 27 cells: three groups $\times t^* \in \{0.5, 1, 4\} \times N \in \{10^2, 10^3, 10^4\}$.

Bias benchmark. The MLE solves $\mu(\hat{t}) = \bar{m}$ with $\mu(t) = \langle C_2 \rangle_t$ and $\bar{m} = N^{-1} \sum_i C_2(R_i)$; expanding μ^{-1} around $\mu(t^*)$ with the cumulant identities of the family ($\mu' = -\frac{1}{2} \text{Var}_t(C_2)$, $\mu'' = \frac{1}{4} \kappa_3(C_2)$) gives the first-order bias b_1 of eq. (10), the analytic column below.

Confidence intervals. The Wald interval is $\hat{t} \pm z_{0.975}/\sqrt{NI(\hat{t})}$ with plug-in information; the profile-likelihood interval is $\{t : 2[\ell(\hat{t}) - \ell(t)] \leq \chi_{1,0.95}^2\}$, where the deviance depends on the data only through the sufficient statistic: $2[\ell(\hat{t}) - \ell(t)] = N[(t - \hat{t})\bar{m} + 2(\ln Z(t) - \ln Z(\hat{t}))]$.

At $t^* = 1$ (bias $\pm$ Monte-Carlo standard error; coverage at nominal 95%, MC resolution $\pm 0.49\%$):

group	N	bias $\widehat{\mathbb{E}[\hat{t}]} - t^*$	analytic b_1 (10)	Wald cov.	profile cov.
U(1)	10^2	$+0.0208 \pm 0.0033$	$+0.0200$	0.950	0.954
	10^3	$+0.0018 \pm 0.0010$	$+0.0020$	0.949	0.949
	10^4	$+0.0002 \pm 0.0003$	$+0.0002$	0.950	0.951
SU(2)	10^2	$+0.0058 \pm 0.0018$	$+0.0067$	0.951	0.949
	10^3	$+0.0002 \pm 0.0006$	$+0.0007$	0.959	0.961
	10^4	$+0.0003 \pm 0.0002$	$+0.0001$	0.951	0.952
SU(3)	10^2	$+0.0028 \pm 0.0011$	$+0.0025$	0.952	0.953
	10^3	$+0.0002 \pm 0.0004$	$+0.0003$	0.957	0.956
	10^4	$+0.0002 \pm 0.0001$	$+0.0000$	0.957	0.956

Findings, across all 27 cells (`mle_bias_ci.json`): (i) the empirical bias is consistent with the $1/N$ law (10) within Monte-Carlo error in 26 of 27 cells — the exception is (SU(3), $t^* = 4$, $N = 10^2$), where $b_1 \approx 0.010$ but the measured bias is -0.003 ± 0.004 , i.e. higher-order terms are still comparable at strong coupling and small N ; (ii) Wald coverage lies in $[0.941, 0.960]$ and profile coverage in $[0.943, 0.961]$, both consistent with the nominal 0.950 at the ± 0.005 Monte-Carlo resolution, with no systematic under-coverage even at $N = 10^2$; (iii) Wald and profile intervals agree in mean length to better than 0.3% everywhere (the log-likelihood is nearly quadratic in t at these sample sizes); (iv) no batch hit the estimator bracket. Bias is negligible relative to the statistical error in every cell ($|b_1| \leq 0.15 \sigma_{\text{CR}}$ at $N = 10^2$ across all cells, falling as $N^{-1/2}$ relative), so the RMSE figures of Sec. B.1 are variance-dominated, as the Cramér–Rao comparison assumed.

C Reproducibility

All computations are deterministic (seed 20260609) and complete in seconds on a laptop. Repository layout (paths relative to repo root):

- `ym2/FORMULAS.md` — every literature formula with citation and verbatim source string, verified against the arXiv LaTeX of [9], [10], [8] (fetched 2026-06-09); the convention discrepancies found ([10] Casimir $\times 2$; additive counterterm) and resolutions.
- `ym2/ym2_flux.py` — U(1) and SU(2): curves, monotonicity, inversion, MLE/CRB demo. Output: `results.json`, `fig_ym2.png/.pdf` (Fig. 1). Runtime ≈ 8 s.
- `ym2/ym2_su3.py` — **new for this paper.** SU(3) (convention asserted at runtime on the irreps 1, 3, $\bar{3}$, 6, 8, 10), plus the n -interval analysis for all three groups: identity (9) by central differences, and the Fisher non-scaling demonstration. Output: `su3_results.json`, `fig_su3.png/.pdf` (Fig. 2). Runtime ≈ 3 s. Run `python3 ym2_su3.py` (no arguments) for the full pipeline; staged execution (`stage1`, `stage2:0.5,1.0`, `stage2:2.0,4.0`, `stage3`, `merge`) produces bit-identical results via per-component RNG streams `default_rng([SEED, stage, index])`.

- `paper3/mle_bias_ci.py` — **new for this revision.** Finite-sample bias and confidence-interval analysis of Sec. B.2: 27 cells (three groups $\times t^* \in \{0.5, 1, 4\} \times N \in \{10^2, 10^3, 10^4\}$), $M = 2000$ batches per cell, per-cell RNG streams `default_rng([SEED, gi, ti, ni])` (so cells can be run in any order or in parallel with bit-identical results). Runs with or without SciPy (a pure-Python Brent root-finder is included). Output: `mle_bias_ci.json`. Runtime ≈ 2 min.
- `paper3/weak_coupling_check.py` — **new for this revision.** Deterministic verification of Appendix A: the constants c_G and the $I(t) = \dim G/(2t^2)$ law on $t \in [10^{-3}, 0.2]$. Output: `weak_coupling_check.json`. Runtime ≈ 20 s.
- Truncation policy: curve/identity computations keep all irreps with relative weight $> e^{-800}$ (truncation error exactly zero in float64); the MLE inner loop uses e^{-120} (relative truncation $< 10^{-50}$, ~ 46 orders below Monte-Carlo noise). For $SU(3)$ the irrep sum runs over a (p, q) grid with $p, q \leq \sqrt{6 \cdot 860/t} + 2$.
- Headline residuals: $|S_{\text{direct}} - S_{\text{exponential-family}}| \leq 3.6 \times 10^{-15}$ ($SU(3)$, 400 grid points); identity (9) to 2.4×10^{-10} (48 combinations); inversion $\leq 9 \times 10^{-14}$ (18 test points); capacity plateau $\frac{1}{2} \dim G$ to $< 10^{-6}$ at $t = 0.05$.

References

- [1] W. T. Trevena, “No-Go Results for a Three-Qubit Entropy Ansatz for Gauge-Coupling Hierarchies, with an Exact Octonionic Rotor Construction in an Appendix,” companion paper (2026), repository `paper/main.tex`.
- [2] W. T. Trevena, “Gauge-Invariant Algebraic Entropy in a Minimal Fermionic Toy: A Single-Copy No-Go for Coupling Encoding,” companion paper (2026), repository `paper2/main.tex`.
- [3] W. T. Trevena, “Fisher Information in Compact $U(1)$ Flux Sectors: Asymptotic Exponential Families, a Crossover Obstruction, and a Topological Zero-Mode Bridge to Four Dimensions,” companion manuscript (2026), repository `paper5/main.tex`.
- [4] A. A. Migdal, “Recursion equations in gauge field theories,” *Sov. Phys. JETP* 42, 413 (1975).
- [5] B. Rusakov, “Loop averages and partition functions in $U(N)$ gauge theory on two-dimensional manifolds,” *Mod. Phys. Lett. A* 5, 693 (1990).
- [6] E. Witten, “On quantum gauge theories in two dimensions,” *Commun. Math. Phys.* 141, 153 (1991).
- [7] S. Cordes, G. Moore, and S. Ramgoolam, “Lectures on 2D Yang–Mills theory, equivariant cohomology and topological field theories,” *Nucl. Phys. B Proc. Suppl.* 41, 184 (1995); arXiv:hep-th/9411210.
- [8] W. Donnelly, “Decomposition of entanglement entropy in lattice gauge theory,” *Phys. Rev. D* 85, 085004 (2012); arXiv:1109.0036.
- [9] W. Donnelly, “Entanglement entropy and nonabelian gauge symmetry,” *Class. Quantum Grav.* 31, 214003 (2014); arXiv:1406.7304.

- [10] A. Gromov and R. A. Santos, “Entanglement entropy in 2D non-abelian pure gauge theory,” *Phys. Lett. B* 737, 60 (2014); arXiv:1403.5035.
- [11] A. Velytsky, “Entanglement entropy in $d+1$ $SU(N)$ gauge theory,” *Phys. Rev. D* 77, 085021 (2008); arXiv:0801.4111.
- [12] H. Casini, M. Huerta, and J. A. Rosabal, “Remarks on entanglement entropy for gauge fields,” *Phys. Rev. D* 89, 085012 (2014); arXiv:1312.1183.
- [13] S. Ghosh, R. M. Soni, and S. P. Trivedi, “On the entanglement entropy for gauge theories,” *JHEP* 09 (2015) 069; arXiv:1501.02593.
- [14] R. M. Soni and S. P. Trivedi, “Entanglement entropy in $(3+1)$ -d free $U(1)$ gauge theory,” *JHEP* 02 (2017) 101; arXiv:1608.00353.
- [15] U. Moitra, R. M. Soni, and S. P. Trivedi, “Entanglement entropy, relative entropy and duality,” *JHEP* 08 (2019) 059; arXiv:1811.06986.
- [16] W. Donnelly and A. C. Wall, “Entanglement entropy of electromagnetic edge modes,” *Phys. Rev. Lett.* 114, 111603 (2015); arXiv:1412.1895.
- [17] W. Donnelly and A. C. Wall, “Geometric entropy and edge modes of the electromagnetic field,” *Phys. Rev. D* 94, 104053 (2016); arXiv:1506.05792.
- [18] H. Casini, M. Huerta, J. M. Magán, and D. Pontello, “On the logarithmic coefficient of the entanglement entropy of a Maxwell field,” *Phys. Rev. D* 101, 065020 (2020); arXiv:1911.00529.
- [19] D. Kabat, “Black hole entropy and entropy of entanglement,” *Nucl. Phys. B* 453, 281 (1995); arXiv:hep-th/9503016.
- [20] Đ. Radičević, “Entanglement in weakly coupled lattice gauge theories,” *JHEP* 04 (2016) 163; arXiv:1509.08478.
- [21] A. Blommaert, T. G. Mertens, and H. Verschelde, “Edge dynamics from the path integral: Maxwell and Yang–Mills,” *JHEP* 11 (2018) 080; arXiv:1804.07585.
- [22] W. Donnelly, S. Timmerman, and N. Valdés-Meller, “Entanglement entropy and the large N expansion of two-dimensional Yang–Mills theory,” *JHEP* 04 (2020) 182; arXiv:1911.09302.
- [23] D. Melnikov, J. T. Oliveira, V. Peixoto, and M. Tenser, “States of 2D Yang–Mills and large-volume entanglement,” arXiv:2603.10171 (2026).
- [24] R. Amorosso, S. Syritsyn, and R. Venugopalan, “Entanglement entropy of a color flux tube in $(1+1)$ D Yang–Mills theory,” arXiv:2411.12818; “Entanglement entropy of a color flux tube in $(2+1)$ D Yang–Mills theory,” *JHEP* 12 (2024) 177; arXiv:2410.00112.
- [25] H. Yao and X.-L. Qi, “Entanglement entropy and entanglement spectrum of the Kitaev model,” *Phys. Rev. Lett.* 105, 080501 (2010); arXiv:1001.1165.
- [26] J. de Boer, J. Järvelä, and E. Keski-Vakkuri, “Aspects of capacity of entanglement,” *Phys. Rev. D* 99, 066012 (2019); arXiv:1807.07357.
- [27] P. Nandy, “Capacity of entanglement in local operators,” *JHEP* 07 (2021) 019; arXiv:2106.00228.

- [28] M. Goldstein and E. Sela, “Symmetry-resolved entanglement in many-body systems,” *Phys. Rev. Lett.* 120, 200602 (2018); arXiv:1711.09418.
- [29] J. C. Xavier, F. C. Alcaraz, and G. Sierra, “Equipartition of the entanglement entropy,” *Phys. Rev. B* 98, 041106(R) (2018); arXiv:1804.06357.
- [30] R. Arias, G. Di Giulio, E. Keski-Vakkuri, and E. Tonni, “Probing RG flows, symmetry resolution and quench dynamics through the capacity of entanglement,” *JHEP* 03 (2023) 175; arXiv:2301.02117.
- [31] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Phys. Rev. Lett.* 72, 3439 (1994).
- [32] R. Camporesi, “Harmonic analysis and propagators on homogeneous spaces,” *Phys. Rep.* 196, 1 (1990).
- [33] Standard sources for $\dim(p, q) = (p+1)(q+1)(p+q+2)/2$ and $C_2(p, q) = (p^2 + q^2 + pq + 3p + 3q)/3$: e.g. A. B. Balantekin et al., lecture notes, arXiv:2109.12087; H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed., Westview Press (1999); Wikipedia, “Clebsch–Gordan coefficients for $SU(3)$ ” (accessed 2026-06-09). Verified to satisfy $C_2(1, 0) = (N^2 - 1)/2N$ and $C_2(1, 1) = N$; asserted at runtime in `ym2_su3.py`.
- [34] H. Barnum, E. Knill, G. Ortiz, and L. Viola, “Generalizations of entanglement based on coherent states and convex sets,” *Phys. Rev. A* 68, 032308 (2003); arXiv:quant-ph/0305023.
- [35] P. Zanardi, “Virtual quantum subsystems,” *Phys. Rev. Lett.* 87, 077901 (2001); arXiv:quant-ph/0103030.